

Postman Test Cases - Request Processor

How to use

- Base URL: `http://localhost:3000`
- Common headers: Content-Type: `application/json`, Idempotency-Key: `<unique>`

1) Create Request (Happy Path)

POST `/requests`

Headers: Idempotency-Key: `happy-1`

Body:

```
{  
  "payload": { "order_id": "A-100", "amount": 4999 }  
}
```

Expected: 202 Accepted, status: pending

Then: GET `/requests/:id` until status is completed

2) Duplicate Request (Same Key + Same Payload)

POST `/requests`

Headers: Idempotency-Key: `dup-1`

Body: { "payload": { "order_id": "A-200" } }

Repeat same request

Expected: 202 Accepted (or 200 if already complete), no duplicate processing

3) Conflict (Same Key + Different Payload)

POST `/requests`

Headers: Idempotency-Key: `conflict-1`

Body: { "payload": { "order_id": "A-300" } }

POST `/requests` with same key and different payload

Body: { "payload": { "order_id": "A-999" } }

Expected: 409 Conflict

4) Retryable Failure

POST `/requests`

Headers: Idempotency-Key: `retry-1`

Body:

```
{  
  "payload": {  
    "order_id": "A-400",  
    "simulate": { "failure_mode": "retryable" }  
  }  
}
```

Expected: 202 Accepted

GET `/requests/:id` shows retrying, attempts increase, final status failed after max attempts

5) Flaky Failure (Fails then Succeeds)

POST `/requests`

Headers: Idempotency-Key: `flaky-1`

Body:

```
{
  "payload": {
    "order_id": "A-500",
    "simulate": { "failure_mode": "flaky", "failures_before_success": 2 }
  }
}
```

Expected: retrying for first 2 attempts, then completed

6) Non-Retryable Failure

POST /requests

Headers: Idempotency-Key: nonretry-1

Body:

```
{
  "payload": {
    "order_id": "A-600",
    "simulate": { "failure_mode": "non_retryable" }
  }
}
```

Expected: status failed, no retry

7) Slow Processing + Cancel

POST /requests

Headers: Idempotency-Key: slow-1

Body:

```
{
  "payload": {
    "order_id": "A-700",
    "simulate": { "delay_seconds": 10 }
  }
}
```

Immediately POST /requests/:id/cancel

Expected: status cancelled

8) Cancel After Completion

Create a normal request, wait until completed, then cancel

POST /requests/:id/cancel

Expected: 409 Conflict

9) Bad Request - Missing Idempotency-Key

POST /requests (no Idempotency-Key header)

Expected: 400 Bad Request

10) Bad Request - Invalid Payload Type

POST /requests

Headers: Idempotency-Key: bad-1

Body: { "payload": "not-an-object" }

Expected: 400 Bad Request

11) Concurrency Test (Same Key)

Send 10 requests concurrently with same Idempotency-Key and same payload

Expected: no duplicates, one record only

12) Status Check

GET /requests/:id

Expected: 200 OK, returns current status and timestamps

Curl Commands (copy/paste)

Happy path

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: happy-1" -d '{"payload":{"order_id":"A-100","amount":4999}}'
```

Check status

```
curl -i http://localhost:3000/requests/<ID>
```

Cancel

```
curl -i -X POST http://localhost:3000/requests/<ID>/cancel
```

Duplicate (same key + same payload)

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: dup-1" -d '{"payload":{"order_id":"A-200"}}'
```

Conflict (same key + different payload)

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: conflict-1" -d '{"payload":{"order_id":"A-300"}}'
```

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: conflict-1" -d '{"payload":{"order_id":"A-999"}}'
```

Retryable failure

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: retry-1" -d '{"payload":{"order_id":"A-400","simulate":{"failure_mode":"retryable"}}}'
```

Flaky failure

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: flaky-1" -d '{"payload":{"order_id":"A-500","simulate":{"failure_mode":"flaky","failures_before_success":2}}}'
```

Non-retryable failure

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H "Idempotency-Key: nonretry-1" -d '{"payload":{"order_id":"A-600","simulate":{"failure_mode":"non_retryable"}}}'
```

Slow processing + cancel

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H  
"Idempotency-Key: slow-1" -d  
'{"payload":{"order_id":"A-700","simulate":{"delay_seconds":10}}}'
```

```
curl -i -X POST http://localhost:3000/requests/<ID>/cancel
```

Missing idempotency key

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -d  
'{"payload":{"order_id":"A-800"}}'
```

Invalid payload type

```
curl -i -X POST http://localhost:3000/requests -H "Content-Type: application/json" -H  
"Idempotency-Key: bad-1" -d '{"payload":"not-an-object"}
```